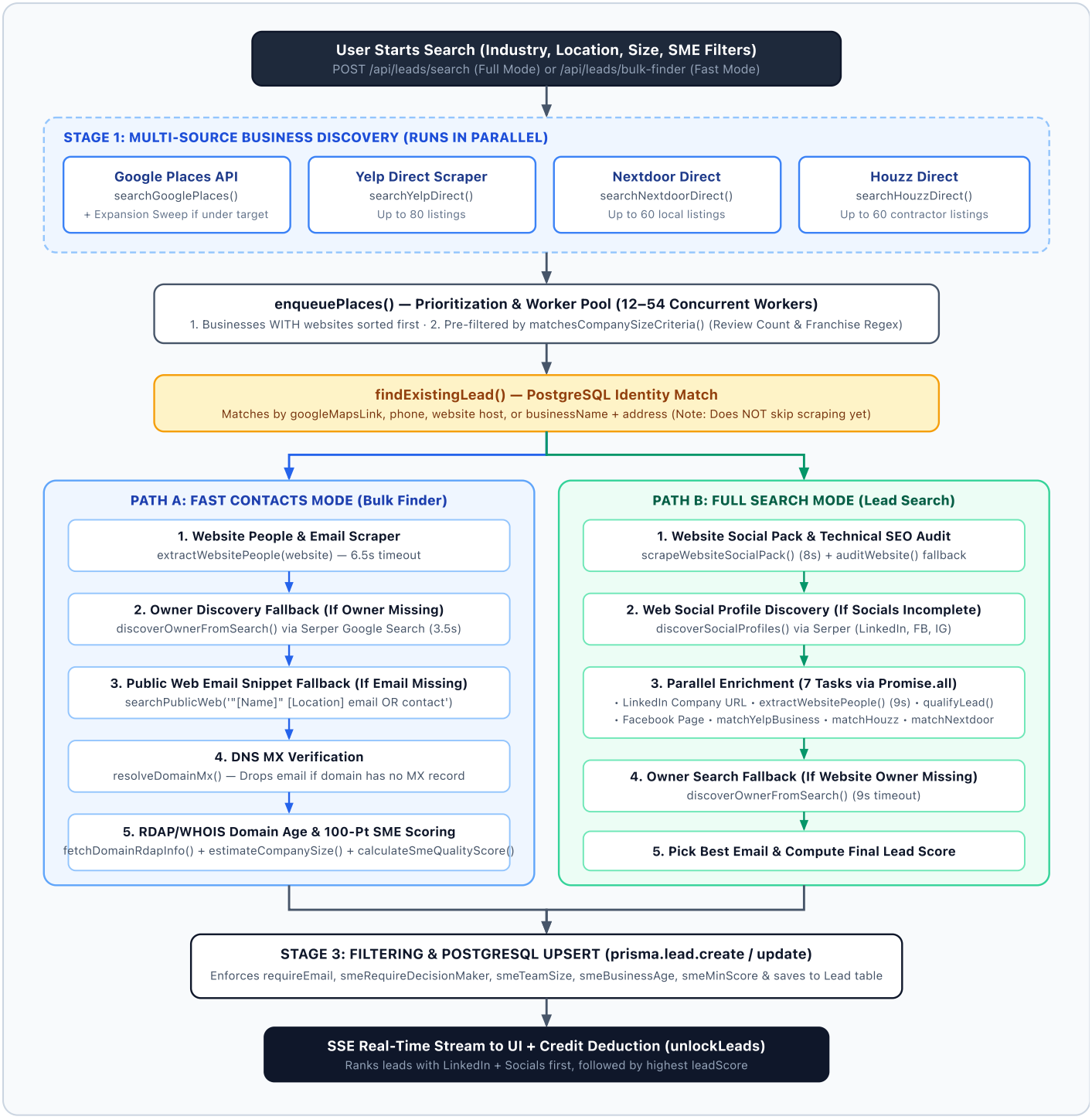
ContractorLeads — Current Lead Finding & Enrichment Architecture

End-to-End Flow in `src/lib/services/lead-pipeline.ts`

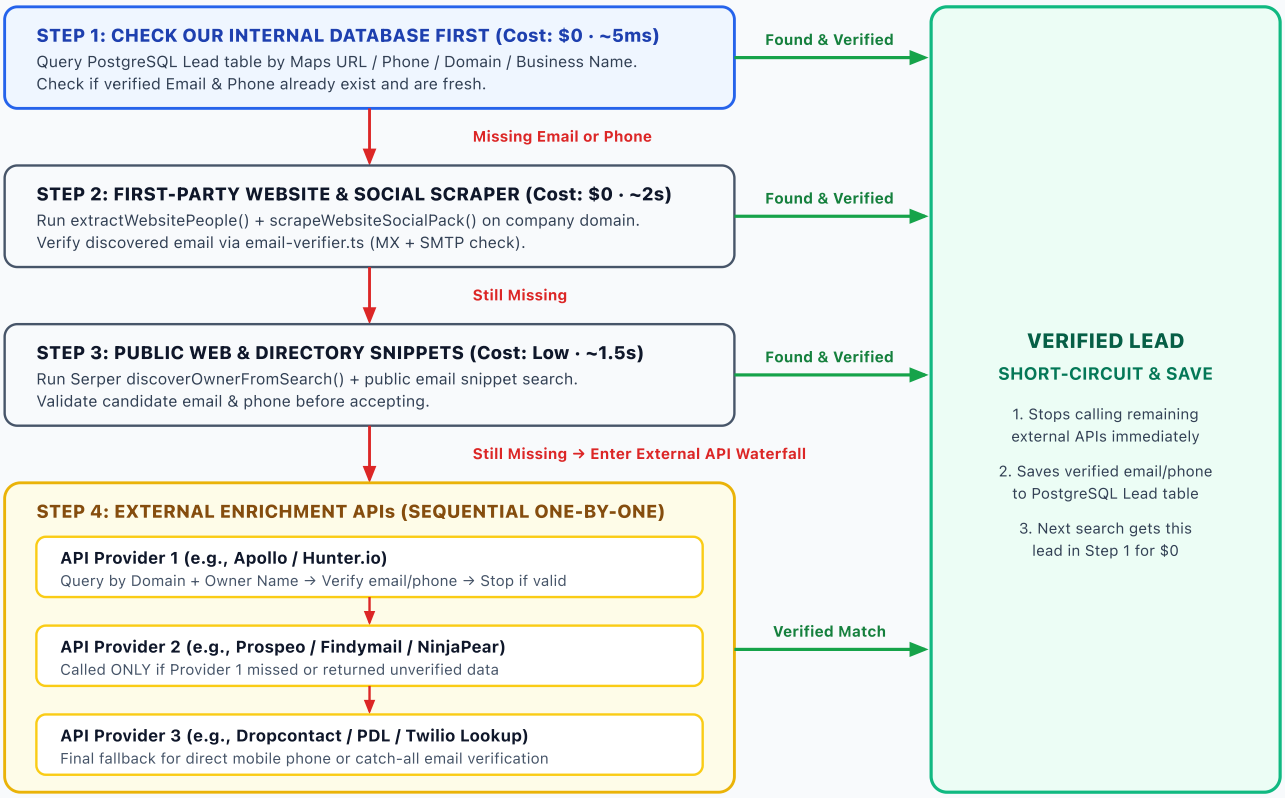
PAGE 1 · CURRENT SYSTEM



Proposed Upgrade — Waterfall Enrichment & API Integration

PAGE 2 · TARGET WATERFALL

Sequential "Stop-on-Success" Pipeline (Internal DB First → Scrapers → External APIs)



CURRENT SYSTEM VS. WATERFALL ENRICHMENT COMPARISON

Stage / Feature	How Our System Works Right Now	How Waterfall Enrichment Works
1. Internal Database Check	Calls Google Places, Yelp, Nextdoor, and Houzz first; even if the lead exists in PostgreSQL, it re-scrapes the website and re-runs Serper searches.	Checks PostgreSQL `Lead` pool first. If verified email & phone exist, returns immediately in ~5ms for $0.
2. Website & Social Scraping	Runs on every discovered place regardless of whether contact info is already known in the database.	Runs **only** when the internal database record is missing a verified email, phone, or decision-maker.
3. External Contact APIs	Only uses Serper Google search snippets (`searchPublicWeb`). `ninjapear.ts` is only used on manual social enrich.	Chains multiple external APIs (Provider 1 → Provider 2 → Provider 3) one by one, stopping as soon as verified data is found.
4. Email & Phone Verification	Only checks domain MX records (`resolveDomainMx`) in Fast Contacts mode; no MX/SMTP check in Full Search mode.	Every candidate email/phone is verified via `email-verifier.ts` before stopping the waterfall. If invalid, moves to next provider.